ALGEBRAIZATION OF A FORMAL ALGEBRAIC SPACE OF RELATIVE DIMENSION AT MOST ONE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 10355-10393, immediately following the statement at 10338-10353.

Editorial note. Five cross-chapter pointers (Cohomology of Schemes, Cohomology, Varieties, Spaces over Fields) plus one same-chapter lemma are recorded in assembly.external_references.

Editorial note. The obstruction group and its vanishing are displayed inside the span.

Allowed standard prerequisites (editorial).

- Obstruction theory for invertible sheaves on thickenings. Cohomology, lemma cohomology-lemma-h1-invertible, cited at proof line 10380. Turns the lifting of an invertible sheaf along the thickening into the vanishing of a class in H^2 of the kernel.
- Vanishing of the obstruction group in dimension at most one. Cohomology, proposition cohomology-proposition-vanishing-Noetherian, cited at proof lines 10385-10386. Kills the obstruction because X_1 is a Noetherian scheme of dimension at most 1.
- Grothendieck's algebraization theorem. Cohomology of Schemes, theorem coherent-theorem-algebraization, cited at proof lines 10389-10391. Produces the projective morphism of schemes X -> S with the compatible system of isomorphisms X_n = S_n x_S X.
- Realising the formal thickening by a scheme. chapter lemma lemma-thickening-scheme, cited in the proof. Lets the formal algebraic spaces X_n be treated as thickenings of the scheme X_1.
- Proper projective and dimension-one facts about X_1. Varieties, lemma varieties-lemma-dim-1-proper-projective and Spaces over Fields, lemma spaces-over-fields-lemma-codim-1-point-in-schematic-locus, cited in the proof. Control the geometry of the one-dimensional fibre so that the cohomological vanishing applies.

Source reference. Original reference 089A: focused-seeds/geometry/S9-spaces-formal-space-proper-reldim-1/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09NU: focused-seeds/geometry/S9-spaces-formal-space-proper-reldim-1/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 02UZ: focused-seeds/geometry/S9-spaces-formal-space-proper-reldim-1/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05ZR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1730–1730. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ADD: focused-seeds/geometry/S9-spaces-formal-space-proper-reldim-1/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A26: focused-seeds/geometry/S9-spaces-formal-space-proper-reldim-1/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a complete local Noetherian ring. Set $S = \mathrm{Spec}(A)$ and $S_n = \mathrm{Spec}(A/\mathfrak{m}^n)$. Consider a commutative diagram*

$$\begin{array}{ccccccc} X_1 & \xrightarrow[i_1]{} & X_2 & \xrightarrow[i_2]{} & X_3 & \longrightarrow & \ldots \\ \downarrow & & \downarrow & & \downarrow & & \\ S_1 & \longrightarrow & S_2 & \longrightarrow & S_3 & \longrightarrow & \ldots \end{array}$$

of algebraic spaces with cartesian squares. If $\dim(X_1) \leq 1$, then there exists a projective morphism of schemes $X \to S$ and isomorphisms $X_n \cong X \times_S S_n$ compatible with i_n.

Proof. By Spaces over Fields, Lemma 0ADD the algebraic space X_1 is a scheme. Hence X_1 is a proper scheme of dimension ≤ 1 over κ. By Varieties, Lemma 0A26 we see that X_1 is H-projective over κ. Let $\mathcal{L}_1$ be an ample invertible sheaf on X_1.

We are going to show that $\mathcal{L}_1$ lifts to a compatible system $\{\mathcal{L}_n\}$ of invertible sheaves on $\{X_n\}$. Observe that X_n is a scheme too by Lemma 05ZR. Recall that $X_1 \to X_n$ induces homeomorphisms of underlying topological spaces. In the rest of the proof we do not distinguish between sheaves on X_n and sheaves on X_1. Suppose, given a lift $\mathcal{L}_n$ to X_n. We consider the exact sequence

$$1 \to (1 + \mathfrak{m}^n \mathcal{O}_{X_{n+1}})^* \to \mathcal{O}_{X_{n+1}}^* \to \mathcal{O}_{X_n}^* \to 1$$

of sheaves on X_{n+1}. The class of $\mathcal{L}_n$ in $H^1(X_n, \mathcal{O}_{X_n}^*)$ (see Cohomology, Lemma 09NU) can be lifted to an element of $H^1(X_{n+1}, \mathcal{O}_{X_{n+1}}^*)$ if and only if the obstruction in $H^2(X_{n+1}, (1 + \mathfrak{m}^n \mathcal{O}_{X_{n+1}})^*)$ is zero. As X_1 is a Noetherian scheme of dimension ≤ 1 this cohomology group vanishes (Cohomology, Proposition 02UZ).

By Grothendieck's algebraization theorem (Cohomology of Schemes, Theorem 089A) we find a projective morphism of schemes $X \to S = \mathrm{Spec}(A)$ and a compatible system of isomorphisms $X_n = S_n \times_S X$. □